

EDUCATION

University of California San Diego	San Diego, CA	Sep 2021 – March 2025
• B.S. in Electrical Engineering, GPA: 3.8 • Coursework: Digital Design, Advanced Digital Design Project, Analog Circuits, Signals, Linear Electronic Systems, Control Systems, Differential Equations, Linear Algebra, Vector Calculus, Linear and Non-linear Optimization, Object-Oriented Programming, Data Structures, Algorithms, Software Development, Machine Learning, Fluid Dynamics		
University of California San Diego	San Diego, CA	March 2025 – June 2026 (Expected)
• M.S. in Electrical Engineering, depth in Intelligent Systems, Robotics & Control		

LANGUAGES AND TECHNOLOGIES

- **Hardware & Systems:** Verilog, SystemVerilog, UVM, RTL Design, Formal Verification, PCB Design, EDA Tools, DFM
- **Software & Firmware:** C/C++, Python, FreeRTOS, Linux, UART, SPI, I2C, CAN Bus, Tcl, MATLAB
- **Tools:** Quartus Prime, Vivado, ModelSim, Allegro, PSpice, Eagle, Oscilloscope, Multimeter, Git

EXPERIENCE

Avionics Lead	SEDS Rocketry at UCSD, San Diego	December 2021 – Present
• Led the 6-member avionics team for <i>Riptide</i>, a bi-propellant, self-landing sounding rocket, focusing on full hardware design, embedded firmware integration, and control system implementation. • Coordinated integration of high-speed peripherals including throttle and gimbal controllers, EPS, and actuators; conducted Hardware-in-the-Loop (HIL) testing to validate control software and embedded system behavior. • Cold-Flow and Hot-fire tested and validated avionics systems at FAR in the Mojave Desert under high-stress conditions. • Designed a modular sensor PCB in Fusion 360, integrating 40+ vehicle sensors such as thermocouples, pressure transducers, GPS, and altimeters using amplifiers, ADCs, and SPI/I²C protocols for high-fidelity data acquisition. • Integrated Teensy microcontroller on sensor board using FreeRTOS in ESP-IDF to transfer sensor data over separate mission-critical and non-critical CAN buses.		
Avionics Electrical Engineering Intern	Aerojet Rocketdyne, Los Angeles	June 2024 – September 2024
• Performed Functional Acceptance Testing on a Valve Driver Assembly PCB, executing rigorous thermal and vibration laboratory tests to ensure compliance with military standards and suitability for space deployment. • Developed the High Voltage Initiator Production Specification, coordinating with cross-functional engineering teams. • Led the Change Control Board (CCB) meeting to finalize production-level specifications and presented to Chief Engineer.		
Manufacturing Controls Intern	Tesla, Fremont	January 2024 - June 2024
• Led the overhaul of the Model Y 1R Seat Assembly line, resolving Ethernet downtime and improving network reliability, saving \$60,000 by mapping and planning the installation of 60 Ethernet devices and 300 I/O modules. • Designed and implemented a Badge-Enabled Safety System PLC Panel for a Factory Machine Shop, automating tool shutdown after set idle time and integrating with 13 machining tools for enhanced safety and operational efficiency. • Wired Seat Function Tester panels in-house for new vehicle model seat testing. Designed a custom PCB adapter module for precision power resistor and implemented dual powering and testing methods for enhanced efficiency.		
Electrical Engineering RnD Intern	HP Inc, Corvallis	June 2023 - December 2023
• Conducted in-depth circuit analysis on 80 returned printhead control boards, diagnosing and repairing for internal use. • Enhanced defect detection by upgrading testing scripts from TCL to Python, adding new failure mode detection cases from Failure Tree Analysis, which improved defect detection rate by 25% and contributed to more efficient Functional Circuit Testing, advancing manufacturing processes. • Reviewed FPGA documentation, including Finite State Machine (FSM) diagrams and I/O configurations, to trace functionality and validate performance during failure analysis, directly contributing to improved board validation processes.		

PROJECTS

- **SHA-256 Bitcoin Hashing in Verilog:** Implemented full SHA-256 algorithm in Verilog and simulated in ModelSim; designed nonce search and difficulty validation logic.
- **Algae-Cleaning Robot (Embedded CV):** Designed custom ESP32-CAM + motor driver PCBs and onboard firmware to support autonomous CV-guided navigation. Implemented PWM, UART, and I2C comms; performed end-to-end PCB design and mechanical integration.